

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	19 July 2026
Team ID	XXXX XXXX XXXX
Project Name	Ucab – Smart Cab Booking Application
Maximum Marks	4 Marks

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story/Sub-Task)
FR-1	User Registration & Authentication	Register via form; login via email/password; JWT-based session
FR-2	Ride Booking	Select pickup & drop location on map; choose cab category
FR-3	Fare & Cab Discovery	Show nearby available cabs with fare estimate and ETA
FR-4	Live Trip Tracking	Track assigned cab's live location on map during the ride
FR-5	Payments	Save payment method; auto-charge on ride completion
FR-6	Booking History	View past rides with fare, route, date, and receipt
FR-7	Add-ons	Apply discount codes; add donation; purchase in-ride refreshments

Non-functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple 3-step booking flow usable without prior training

NFR No.	Non-Functional Requirement	Description
NFR-2	Security	JWT authentication, hashed passwords (bcrypt), HTTPS, input validation
NFR-3	Reliability	Booking and payment operations remain consistent even on network retries
NFR-4	Performance	Cab search and fare estimate respond within 2 seconds
NFR-5	Availability	Backend APIs designed for high uptime with MongoDB Atlas managed hosting
NFR-6	Scalability	Stateless Node.js/Express API layer that can scale horizontally

Solution Requirements — Ucab

FUNCTIONAL REQUIREMENTS	NON-FUNCTIONAL REQUIREMENTS
FR-1 User Registration & Authentication Register / login with JWT-based sessions	NFR-1 Usability Simple 3-step booking flow
FR-2 Ride Booking Select pickup/drop on map, choose cab type	NFR-2 Security JWT, bcrypt hashing, HTTPS, validation
FR-3 Fare & Cab Discovery Nearby cabs with fare estimate & ETA	NFR-3 Reliability Consistent booking/payment on retries
FR-4 Live Trip Tracking Track assigned cab's live location	NFR-4 Performance Cab search & fare respond within 2s
FR-5 Payments Save method; auto-charge on completion	NFR-5 Availability High uptime via MongoDB Atlas
FR-6 Booking History View past rides, fare, route, receipts	NFR-6 Scalability Stateless API layer scales horizontally
FR-7 Add-ons Discount codes, donation, refreshments	

Functional requirements define WHAT Ucab does; non-functional requirements define HOW WELL it does it.